

COMPUTATIONAL PHYSICS

EXERCISES FOR CHAPTER 2

Exercise 2.7: Catalan numbers

The Catalan numbers C_n are a sequence of integers 1, 1, 2, 5, 14, 42, 132... that play an important role in quantum mechanics and the theory of disordered systems. (They were central to Eugene Wigner's proof of the so-called semicircle law.) They are given by

$$C_0 = 1, \quad C_{n+1} = \frac{4n+2}{n+2} C_n.$$

Write a program that prints in increasing order all Catalan numbers less than or equal to one billion.